



S⁴D: Style and Structure Similarity for Site Domains

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Περίληψη

Τα τελευταία χρόνια, οι ιστότοποι έχουν γίνει πιο πολύπλοκοι και ποικιλόμορφοι ως προς τον σχεδιασμό και τη δομή τους. Παρά την ποικιλομορφία αυτή, οι ιστότοποι που βρίσκονται στον ίδιο θεματικό τομέα ή εξυπηρετούν παρόμοιους σκοπούς συχνά μοιράζονται κοινά παρουσιαστικά και δομικά μοτίβα. Η αναγνώριση και η σύγκριση αυτών των μοτίβων σε διαφορετικούς τομείς ιστότοπων είναι πολύτιμη για την επίτευξη αποτελεσμάτων ως προς την ταξινόμηση ιστοσελίδων, την αυτοματοποιημένη ανάλυση διεπαφών χρήστη (UIs) και την αποτύπωση των θεματικών τους τομέων. Ωστόσο, η δομή και η οπτική παρουσίαση των περιεχομένων ιστού είναι συχνά μη τυποποιημένη και ασυνεπής, αυξάνοντας τη δυσκολία τέτοιων αναλύσεων. Σε αυτήν την πτυχιακή εργασία, παρουσιάζεται το **S⁴D**, ένα σύστημα που αυτοματοποιεί τη διαδικασία εξαγωγής, μοντελοποίησης και σύγκρισης των παρουσιαστικών και δομικών ομοιοτήτων διαφορετικών ιστοτόπων. Το σύστημα αναλύει ένα σύνολο ιστοσελίδων μέσω τεχνικών αυτοματοποιημένης περιήγησης, μοντελοποιώντας τα CSS χαρακτηριστικά των στοιχείων τους και οργανώνοντάς τα με τρόπο που επιτρέπει την περαιτέρω ανάλυση. Συγκεκριμένα, μετατρέποντας τα στοιχεία σελίδας σε διανυσματικές αναπαραστάσεις και εφαρμόζοντας τεχνικές μη-επιβλεπόμενης συσταδοποίησης, το σύστημα εντοπίζει ουσιαστικές συστάδες των στοιχείων βάσει των οπτικών τους χαρακτηριστικών. Χρησιμοποιώντας αυτήν την συσταδοποιημένη αναπαράσταση, το σύστημα μοντελοποιεί τα παραγόμενα δεδομένα για κάθε ιστότοπο, εξάγει μοτίβα διάταξης στοιχείων και υπολογίζει την ομοιότητα μεταξύ τους χρησιμοποιώντας διαφορετικές μετρικές απόστασης. Τέλος, το σύστημα παρουσιάζει έναν πίνακα ομοιότητας μεταξύ των ιστοτόπων, προσφέροντας πληροφορίες σχετικά με τη δομική και παρουσιαστική ομοιότητα των ιστοσελίδων τους.

Abstract

In recent years, websites have become increasingly complex and diverse in design and structure. Despite this diversity, websites belonging to the same domain or serving similar purposes frequently exhibit similar stylistic and structural characteristics. Identifying and comparing these patterns across different site domains is valuable for tasks such as web classification, automated user interface (UI) analysis, and domain topic labeling. However, the structure and visual presentation of web content is often unstandardized and inconsistent, making such analysis a challenging task. This thesis introduces **S⁴D**, a system that automates the process of extracting, modeling, and comparing the style and structure of websites across different domains. The system parses a set of webpages using browser automation techniques, modeling CSS-related properties of web elements, and structuring them in a way that enables further analysis. By transforming page elements into vector representations and applying unsupervised clustering techniques, the system identifies meaningful groupings of visual and structural components. Using this clustered representation, the system exploits the resulting data modeling of each domain, extracting high-level patterns of element arrangement, and computes the inter-domain similarity using multiple distance metrics. Finally, the system presents a similarity matrix between domains, providing insight into the structural and stylistic resemblance of their webpages.

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Chapter 1

Introduction

This thesis presents **S⁴D**, a system for analyzing and detecting similarities between websites. **S⁴D** is a multi-layered system that aims to facilitate the modeling and analysis of websites' visual and structural similarity. Through a variety of functionalities and a Graphical User Interface (GUI), it is able to yield meaningful insight into the websites' visual and structural similarity analysis process, ultimately generating a similarity matrix for all input webpages.

1.1 Problem Statement

The current landscape of web analysis tools focuses primarily on content-based analysis, often overlooking the significance of visual design patterns and structural similarities between websites. In the era of rapidly expanding web content, understanding the structural similarity between websites has become increasingly important for various applications including web mining, content analysis, and automated web processing. However, traditional approaches face several critical challenges.

1.1.1 Web Structure Analysis Challenges

Modern web structure analysis presents several unique technical challenges that our S⁴D system specifically addresses:

- **DOM Complexity and Recursive Processing.** Modern websites contain deeply nested DOM structures with thousands of elements, requiring recursive traversal algorithms to accurately capture hierarchical relationships. The sheer depth and breadth of contemporary HTML trees make consistent and scalable structural analysis a non-trivial engineering challenge.
- **CSS Style Diversity and Standardization.** The variety of CSS styling approaches across websites creates inconsistency in structural representation, making cross-site comparisons particularly challenging. Different sites express the same visual properties through differing units, formats, and conventions, requiring unified normalization before any meaningful comparison can take place.
- **Attribute Extraction and Feature Engineering.** Extracting meaningful features from HTML elements involves processing hundreds of CSS properties per element. Identifying which properties carry structural significance — and how to normalize them for comparison — is central to building a reliable similarity model.
- **Scalability and Performance Optimization.** Processing large numbers of websites demands efficient algorithms and robust memory management. As the number of analyzed domains grows,

the computational overhead of feature extraction and comparison increases substantially, requiring careful architectural choices to ensure reliable performance at scale.

- **Multi-dimensional Similarity Computation.** Defining meaningful similarity measures for web structures involves multiple algorithmic challenges, since web pages differ across visual, structural, and positional dimensions simultaneously. No single metric fully captures the richness of inter-site similarity, making combined approaches necessary.
- **Clustering Parameter Optimization.** Achieving optimal clustering results requires careful parameter tuning, as different web datasets exhibit varying densities and structural patterns. Selecting appropriate parameters without overfitting to a specific domain is a non-trivial problem that demands principled search strategies.
- **Cross-Domain Pattern Recognition.** Identifying design trends across different domains requires sophisticated comparison frameworks capable of abstracting away domain-specific content while preserving structural and stylistic signals. This becomes especially challenging when the overall design language varies significantly between domains.

1.1.2 Research Motivation

The need for automated web structural analysis stems from several key factors:

- **Content Management:** Organizations need tools to understand and manage large collections of web content based on structural and visual characteristics.
- **Design Pattern Analysis:** Web developers and designers benefit from understanding common structural patterns and visual trends across websites in their domain.
- **Accessibility Evaluation:** Automated structural analysis can help identify accessibility issues and design inconsistencies across multiple web properties.
- **Web Analytics Enhancement:** Structural similarity analysis provides deeper insights into website categorization and user experience patterns beyond traditional content-based metrics.

1.2 Our Methodology and Contributions

S⁴D addresses the challenges outlined above through a cohesive set of methods and components. To handle DOM complexity, the system uses unique ID generation and hierarchical tree processing to manage parent-child relationships across depth levels 1–10 while maintaining structural integrity throughout analysis. For CSS diversity, sophisticated parsing and normalization pipelines process pixel values (**px**), percentages (**%**), and relative units in a unified manner, enabling consistent cross-site comparisons.

Attribute extraction is guided by selective filtering focused on structurally significant properties such as **font-size**, **text-align**, **line-height**, and **cursor**, with dedicated handling for color space conversions and unit standardization. Scalability is achieved through domain-based processing modes, ChromeDriver optimization, and parallel data processing, enabling reliable batch analysis across large website collections.

For similarity computation, the system combines HDBSCAN clustering with DBCV (Density-Based Cluster Validation) scoring [6], cosine similarity matrices, and role vector analysis to jointly capture structural, visual, and positional relationships between elements. Clustering parameters are tuned via grid search over cluster size ranges and selection epsilon values, with automatic outlier reassignment to the

nearest cluster centroid using Euclidean distance. Cross-domain analysis is enabled through domain-level aggregation, transition matrices between design elements, and the use of the cosine similarity metric.

This thesis makes the following key contributions:

- **Novel feature extraction methodology:** Development of comprehensive techniques for extracting both structural and visual features from websites
- **Advanced similarity metrics:** Introduction of new metrics specifically designed for website comparison
- **Integrated clustering approach:** Implementation of clustering algorithms optimized for website analysis
- **Comprehensive evaluation framework:** Establishment of evaluation metrics and datasets for assessing website similarity systems
- **Practical applications:** Demonstration of real-world applications in web design analysis and automated categorization

1.3 Research Objectives

To tackle the challenges raised above, the primary objectives of this research are:

1. To develop a robust system for extracting and analyzing structural and visual features from websites
2. To create effective similarity metrics that can accurately measure relationships between different websites
3. To implement clustering algorithms that can group websites based on their design and structural characteristics
4. To provide a comprehensive comparison framework for website analysis across different domains
5. To validate the effectiveness of the proposed system through extensive testing and evaluation

1.4 Thesis Structure

The remainder of this thesis is organized as follows:

- **Chapter 2: Literature Review** presents a comprehensive review of existing approaches to website analysis, similarity measurement, and clustering techniques relevant to web data.
- **Chapter 3: Methodology** describes the proposed S⁴D system architecture, including feature extraction methods, similarity metrics, and clustering algorithms.
- **Chapter 4: Implementation** details the technical implementation of the system, including data structures, algorithms, and software architecture decisions.
- **Chapter 5: Experimental Design** outlines the experimental methodology, datasets used, and evaluation metrics employed to assess the system's performance.
- **Chapter 6: Results and Analysis** presents the experimental results, performance analysis, and comparison with existing approaches.

- **Chapter 7: Discussion** provides an in-depth discussion of the findings, limitations, and implications of the research.
- **Chapter 8: Conclusion and Future Work** summarizes the key findings, contributions, and suggests directions for future research.

Chapter 2

Related Work

This chapter reviews research related to the structural and visual analysis of web pages, highlighting prior work in web parsing, information extraction, and representation learning. The discussion emphasizes how these approaches inform, contrast with, and complement the S⁴D system. The review is structured into three major areas: (1) structural understanding of web documents, (2) information extraction and representation approaches, and (3) similarity and clustering methods for web analysis. Finally, we compare S⁴D with prior systems to highlight its unique contributions.

2.1 Structural Understanding of Web Documents

One of the earliest challenges in web analysis has been the ability to combine textual semantics with structural information. Chen et al. [1] introduced **WebSRC**, a large-scale dataset for structural reading comprehension. Their work demonstrates that effective web understanding requires not only semantic interpretation but also the hierarchical organization of HTML elements. WebSRC provides 400K question-answer pairs across 6.4K web pages, combining HTML source code, screenshots, and metadata. While WebSRC focuses on question answering, its emphasis on DOM-aware structural comprehension is relevant to S⁴D’s structural modeling. Unlike WebSRC, however, S⁴D does not address query answering but instead targets structural and stylistic similarity across domains.

Building on this idea of structural representation, Deng et al. [2] proposed **DOM-LM**, a transformer-based model that encodes both textual content and DOM structure to learn generalizable representations of web pages. Their approach addresses the shortcomings of methods that rely solely on visual rendering or plain-text representations. DOM-LM achieves strong generalization in few-shot and zero-shot tasks such as attribute extraction and information retrieval. In contrast, S⁴D applies structural encoding not for semantic extraction but to identify recurring patterns of element arrangement across domains, enabling inter-domain similarity assessment.

2.2 Information Extraction and Representation

The extraction of structured information from web pages has been another central research theme. Gulhane et al. [3] developed **Vertex**, a wrapper induction system deployed at Yahoo! for large-scale record extraction. Vertex introduced algorithms for grouping similar pages, generating XPath-based extraction rules, and adapting to structural changes. It achieved web-scale deployment, extracting over 250 million records from more than 200 websites. Vertex demonstrates the scalability of web information extraction but is tailored to structured record retrieval. By contrast, S⁴D does not generate wrappers or extract database-like content; it focuses instead on comparing visual and structural layouts across entire domains.

More recently, Lockard et al. [4] proposed **ZeroShotCeres**, which tackles the challenge of zero-shot relation extraction from semi-structured web pages. Their model employs graph neural networks to capture both textual semantics and visual cues such as layout, font size, and color. This enables generalization to previously unseen templates without the need for template-specific training data. ZeroShotCeres highlights the importance of structural and visual cues for generalizable web analysis. While their focus is relation extraction, S⁴D instead leverages structural and positional properties to compute domain-level similarity, not entity-level relations.

2.3 Similarity and Clustering in Web Analysis

Similarity measurement and clustering techniques have been applied in web analysis to group documents, identify templates, and extract structure. Traditional approaches often prioritize textual similarity, but recent advances emphasize structural and visual dimensions. For example, graph-based models such as ZeroShotCeres [4] highlight the utility of representing web pages as graphs where nodes correspond to DOM elements and edges represent structural or semantic relationships.

The S⁴D system extends these ideas by combining multiple similarity dimensions—structural and positional—and introducing role vector similarity to capture the functional roles of elements across domains. For clustering, S⁴D adopts HDBSCAN [5], which automatically determines the number of clusters, handles noise, and reveals hierarchical patterns. This makes it particularly suited to web data, which is both high-dimensional and heterogeneous. Unlike prior work, S⁴D applies clustering at the element level within pages and then aggregates results at the domain level, enabling the analysis of cross-domain stylistic and structural similarity.

2.4 Comparison with Prior Work

The reviewed works demonstrate significant progress in web analysis, yet they address different objectives from S⁴D:

- **WebSRC** [1]: Focuses on question answering over structured web data, highlighting the need for structural comprehension. S⁴D instead uses structure for cross-domain similarity analysis.
- **DOM-LM** [2]: Learns generalizable document embeddings from DOM+text representations. S⁴D leverages DOM+CSS for unsupervised clustering rather than representation learning.
- **Vertex** [3]: Extracts structured records at web scale via wrapper induction. S⁴D does not extract records but compares layout and style across domains.
- **ZeroShotCeres** [4]: Performs zero-shot relation extraction using structural and visual features. S⁴D uses similar structural and positional cues but for similarity computation and domain-level clustering.

2.5 Research Gaps and Opportunities

The literature reveals several research gaps that S⁴D addresses:

- **Cross-domain similarity**: None of the reviewed systems compute similarity matrices across domains, which is central to S⁴D.
- **Multi-dimensional analysis**: Existing work typically emphasizes content or relations; S⁴D integrates structural and positional dimensions.

- **Clustering across stylistic patterns:** Prior systems rarely apply clustering to identify design-level commonalities across domains. S⁴D introduces a multi-level clustering approach for this purpose.
- **Scalability to modern web design:** Systems like Vertex demonstrate scale for extraction tasks, but scalable cross-domain similarity analysis remains underexplored.

2.6 Summary

This chapter has reviewed four representative systems—WebSRC, DOM-LM, Vertex, and ZeroShotCeres—that address different aspects of structural web understanding, information extraction, and similarity-based analysis. Each system contributes important ideas relevant to S⁴D, yet none directly targets cross-domain visual and structural similarity measurement via unsupervised clustering. The gaps identified above motivate the design choices presented in the following chapter.

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